

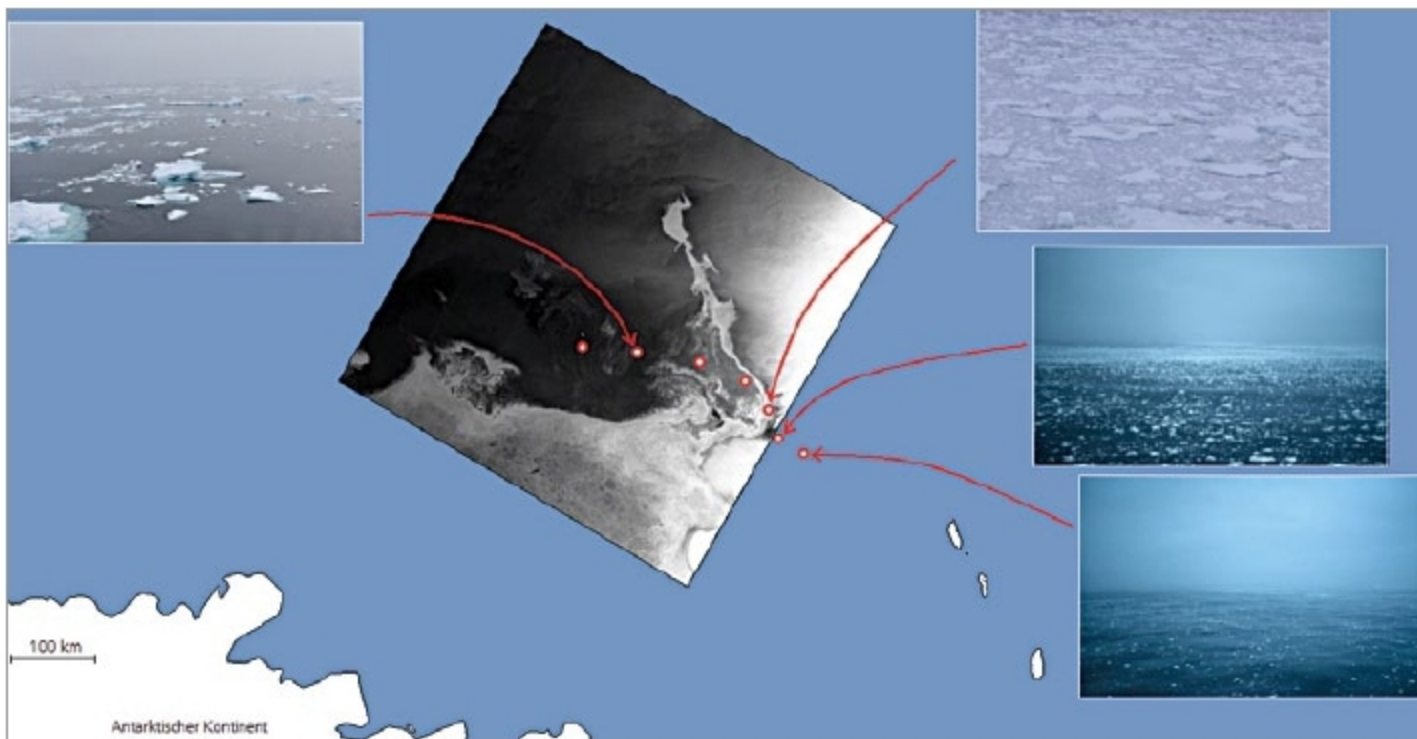


Fig. 2: Synthetic aperture radar and GNSS applied to the fields of geo information sciences (Credit: www.dlr.de)

Combining multiple interferometric sets of images acquired at different times is called differential interferometry, and it can be used to extract information about the deformations suffered by objects over time. Examples of this can be subsidence studies in urban areas, stability in open-pit mines and a building's thermal expansion and contraction, volcanoes and earthquakes, among many others.

If many images of the same area are taken from different angles, tomographic techniques can be applied. SAR tomography can be used to see the inside of a forest or a glacier by sending signals to it from a satellite. This can be used, for instance, to study the amount of biomass on Earth or the inner ice layers in the North Pole.

Location-based services by satellites. This technique includes personalised marketing offers delivered to the user of the smartphone. Offers are based both on personal preferences of the user as well as on the location where the potential customer finds himself, at the moment when the offers are placed. It provides the user with information and the ability to advertise about nearby businesses and services.

This technique is also used by tourists and hikers to locate amusement parks, museums and commercial centres, among others.

The technique can provide precise time and synchronise ground-based

clocks to monitor the energy flow within their own power grids. The high quality of time synchronisation provides better services for the transportation and distribution of energy. Signals provided by navigation satellites may also be utilised to measure the changes occurring in the Earth's magnetic field caused by ionosphere storms. The research concerning the impact of ionosphere storms greatly expands human knowledge on ionosphere dynamics related to the activities of the Sun.

This is an invaluable asset for coastguards and border control authorities, as it ensures faster search and rescue operations, saving more lives. In such situations, the SAR transmitter sends an emergency positioning signal that is received by the satellites, allowing for quick and accurate emergency response. It helps respond to distress signals in case of an emergency. Marine seismic exploration is benefitting from surveys of new sites and identification of geomorphologic risks in drilling activities.

Location-based services by satellites help improve atmospheric measurements, including level of water vapour in the air, which is crucial information for more accurate weather forecasting and a better understanding of climate change. It also helps study the ionosphere and space weather, earthquakes' predictor events, and better understand the movement of animal populations.

Highly accurate and reliable signals play an important role in fleet management, delivering the vehicles' positioning on a map in aviation, maritime, rail and even pedestrian traffic.

Location-based services by satellites assist in monitoring from a structure's planning to its ongoing maintenance and surveillance. The telecom industry uses timing signals for network management, calibration of frequency references, and time-tagging-certified

timestamps that are necessary for electronic banking, financial transactions, e-commerce, and quality assurance systems and services.

Why we need satellite navigation

To save some power, devices switch on GNSS chipset periodically and keep it switched off most of the time. This is a good approach if you do not need to know the location of your device often. A GNSS chipset takes a while to come up with a good solution because it needs to collect several samples. Once it has done that, it continues to deliver good-quality results unless you reach your destination. But if you periodically turn off the chipset, you lose every time the information you had gathered before, and you cannot expect to have good results at all.

GNSSes are becoming more and more important. Not only are these becoming relevant for the economy, these are having a huge impact on the everyday lives of human beings. The aforesaid solutions serve a number of purposes related to transportation or positioning. Using multiple systems also promises to make satellite navigation much faster; if more satellites are in view, initial delay from satellites in downloading data it needs to start calculating your position is reduced. Although, it is still difficult to get signals while working under bridges and in tunnels. **EFY**